Optimization Algorithm Comparison for Material Selection Artificial Neural Networks

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***Abstract*—Substrate warpage has always been a crucial problem in the field of chip packaging. Excessive packaging warpage will lead to delamination, crack and other reliability problems. At the same time, with the development of packaging technology, the existing chip design has become more complex and sophisticated, including more kinds of materials and processes. Among them, different variables are coupled and affect each other, which also puts higher requirements for warpage control. Nowadays, as machine learning and artificial intelligence technologies continue to evolve, it also brings new innovative power to other fields. In the field of packaging, many researches have applied machine learning algorithm to achieve structural design, formula optimization, material selection, and also achieved good results. We noticed that most of the studies focus on achieving more accurate models through different algorithms, and the objective optimization is mostly achieved through genetic algorithms after model construction, and there is a lack of sufficient discussion on the selection of different optimization algorithms. Therefore, we try to compare different optimization algorithms for parameter optimization of our trained machine learning models, and analyze the results.**

In this paper, we constructed a flip chip ball grid array (FCBGA) package structure based on experiments, which used the coefficients of thermal expansion (CTE) and elastic modulus (E) of lid, thermal interface material (Tim), underfill, adhesive and substrate (Sub) as variables. Each material was simplified to an isotropic material and warping simulation analysis of the warming process was performed. We constructed a simulation dataset containing 64 data points by orthogonal experimental design within the reasonable material range of the material parameters. A fully connected neural network (FCNN) containing three hidden layers was trained using the material parameters as input indexes and the substrate warpage values as output indexes, and its R-square index score was verified to be 0.9998 based on the test dataset. Based on this model, we have implemented material parameter combination optimization by three algorithms, genetic algorithm (GA), Broyden—Fletcher—Goldfarb—Shanno (BFGS) method, and simulated annealing (SA), and achieved better results. The results are verified by finite element analysis (FEA) simulation.

Keywords—Warpage, Finite element analysis (FEA), Artificial Neural Network (ANN), Optimization Algorithm

I. INTRODUCTION

In recent years, with the continuous development of packaging technology, the materials contained in a packaging structure have become increasingly complex, making it very difficult for us to improve packaging reliability by modifying the properties of the materials [1]. In our practical work, when adjusting one material, it often leads to new problems with other materials in the packaging instead. Different materials interact with each other, and the interaction effect is very obvious. Warpage has been one of the important causes of various failures, is also a comprehensive manifestation of the CTE mismatch of all materials in electronic devices during temperature changes [2]. Therefore, we wonder if we can use machine learning techniques to select packaging materials within the appropriate range, so as to achieve the purpose of improving the warpage, stress or other indicators. It can not only improve the overall reliability of the packaging, but also avoid analyzing the complex interaction effects of various materials in electronic packaging.

And in existing research, there has been many works that use machine learning to optimize packaging design [3-7]. In our study, based on the previous work [8], we first constructed a FCBGA package simulation model in Abaqus software according to the actual samples. Then the CTE and modulus of five materials lid, adhesive, Tim, underfill, and Sub, were selected as variables, and three or four points were chosen within a reasonable range of material parameters for orthogonal experimental design to construct a training dataset containing 64 data points. And we also construct a test dataset containing ten data. Based on the above data we constructed a fully connected neural network and evaluated it using R-squared (R2) decision coefficients, mean squared error (MSE) and mean absolute error (MAE). After evaluating our constructed artificial neural network, we optimized the model by three optimization methods, BFGS algorithm, genetic algorithm and simulated annealing algorithm. After the validation of the results of the three algorithms by FEA, the effects and conclusions of the algorithms are analyzed and discussed.

II. MODELLING

A. Finite Element Modelling

The standard FCBGA with Lid structure is shown in Fig. 1, for simplicity we regard underfill and solder as the same material UF. And since the mechanical properties of the silicon die are basically unchanged, we select the CTE and E of the rest of five materials as variables for machine learning modelling.

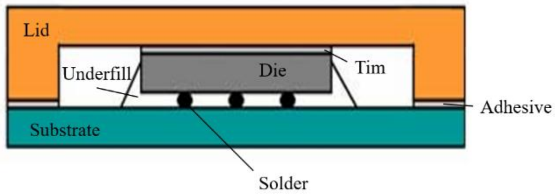

Fig. 1. FCBGA with lid package structure

A 1/4 symmetric FCBGA with lid package finite element model was constructed using Abaqus software, the size of substrate is 25×25 mm^2, the material was set to be an isotropic material, the temperature load setting from 25°C to 150°C and all elements type using C3D8R. The FEM is shown in Fig. 2(a). The model warpage values are extracted by the z direction displacement values on the diagonal path of the baseplate bottom surface as shown in Fig. 2(b).

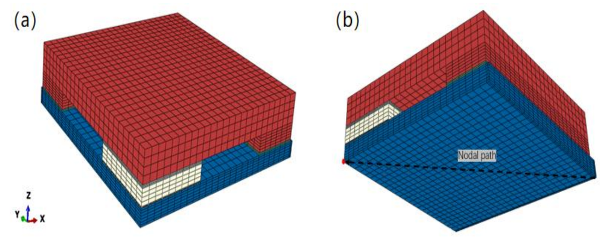

Fig. 2. (a) Quarter symmetric FEA model of flip-chip package, (b) the nodal path for extracting warpage at the bottom of substrate.

B. *Orthogonal Experimental Design*

In our study, ten mechanical property parameter variables were selected according to the FCBGA package model as respectively Lid_CTE, Lid_E, Tim_CTE, Tim_E, Underfill_CTE, Underfill_E, Adhesive_CTE, Adhesive_E, Substrate_CTE, Substrate_E. We took 3 or 4 values at equal intervals within the proper range of all material parameters as variable values for training data, and perform normalization and orthogonal experimental design. Based on the results of orthogonal experimental design, we obtained 64 parameter combinations to form our training data, and we also constructed a test dataset containing 10 data points. The normalized value points of the material, the first ten training data, and the test dataset are shown in Tables I, II, and III.

TABLE I. NORMALIZATION RANGE TABLE

Features	Normalized value point
Lid_CTE	[0, 0.333, 0.667, 1]
Lid_E	[0, 0.5, 1]
Adhesive_CTE	[0, 0.5, 1]
Adhesive_E	[0, 0.5, 1]
Tim_CTE	[0, 0.5, 1]
Tim_E	[0, 0.5, 1]
Underfill_CTE	[0, 0.5, 1]
Underfill_E	[0, 0.5, 1]
Sub_CTE	[0, 0.333, 0.667, 1]
Sub_E	[0, 0.333, 0.667, 1]

TABLE II. FIRST TEN TRAINING DATASETS

Case number	Input features	Warpage (um)
1	[1, 0, 0, 0, 0, 0, 0, 0, 1, 1]	32.71
2	[0.667, 0.5, 0.5, 0, 0, 0, 0, 0.5, 1, 0.333]	31.06
3	[0.667, 0, 0, 0.5, 0, 1, 0, 0.5, 0, 0.667]	14.28
4	[0, 0.5, 0, 1, 0, 0, 0.5, 1, 0.333, 0]	19.52
5	[0.667, 0.5, 0, 0, 0, 1, 0.5, 0, 0.333, 0.667]	19.47
6	[0.333, 0, 0, 1, 0, 0.5, 0, 1, 1, 0.333]	30.35
7	[0, 0, 0, 1, 0.5, 0.5, 1, 0, 0.333, 1]	20.55
8	[1, 0, 0, 0.5, 1, 1, 0.5, 0, 0.667, 0]	22.71
9	[1, 0.5, 0, 0.5, 0, 0.5, 0, 0, 0.667, 0.667]	25.25
10	[0.333, 1, 0, 1, 0.5, 1, 0.5, 0.5, 1, 1]	31.25

TABLE III. TEST DATASETS

Case number	Input features	Warpage (um)
1	[0.1, 0.09, 0.09, 0.1, 0.09, 0.09, 0.09, 0.09, 0.09, 0.09]	15.50
2	[0.17, 0.18, 0.18, 0.2, 0.18, 0.18, 0.18, 0.18, 0.18, 0.18]	17.09
3	[0.27, 0.27, 0.28, 0.25, 0.28, 0.28, 0.28, 0.27, 0.28, 0.27]	18.70
4	[0.37, 0.36, 0.37, 0.35, 0.37, 0.36, 0.37, 0.36, 0.37, 0.36]	20.32
5	[0.47, 0.45, 0.46, 0.45, 0.46, 0.46, 0.46, 0.45, 0.46, 0.45]	21.94
6	[0.53, 0.55, 0.55, 0.55, 0.55, 0.54, 0.55, 0.55, 0.54, 0.55]	23.43
7	[0.63, 0.64, 0.64, 0.65, 0.63, 0.64, 0.64, 0.64, 0.63, 0.64]	25.06
8	[0.73, 0.73, 0.73, 0.75, 0.73, 0.73, 0.73, 0.73, 0.72, 0.73]	26.69
9	[0.83, 0.82, 0.82, 0.8, 0.82, 0.82, 0.82, 0.82, 0.82, 0.82]	28.33
10	[0.9, 0.91, 0.91, 0.9, 0.91, 0.91, 0.91, 0.91, 0.91, 0.91]	29.96

C. *Optimize algorithm*

The BFGS method is a Quasi—Newton method. Newton's method is a process of iteratively finding the stagnation point (extremum) of a function to achieve optimization, starting from a random point function value and continuously approaching the stagnation point based on the results of the first and second derivatives. The iterative formula for Newton's method for a univariate function was shown in (1). However, in practical problems such as our ANN model, it is the process of optimizing multivariate functions. Here, it is necessary to obtain the second derivative of the multivariate function, which includes various partial derivatives of the variables of the function. Taking the function $f = f(x, y, z)$as an example, the second derivative is written in matrix form as shown in (2), and the matrix H is called the Hessian matrix. As the number of variables increases, it becomes more and more difficult to solve for the Hessian matrix, and the Hessian matrix needs to be solved again at each iteration. To avoid this problem, the BFGS method iteratively approximates the Hessian matrix using (3), where $B_K = H_k^{-1}$, $y_k = f'(x_{k+1}) - f'(x_k)$, $S_k = x_{k+1} - x_k$. Compared with the genetic algorithm and simulated annealing algorithm we will use later, BFGS is a numerical solution method based on function derivatives and iterative fast finding of function stagnation points. But the problem with the BFGS method is that if the function itself is very complex, the result of solving the function is easily converge to a local optimum rather than a global optimum, because the local optimum itself is also a stationary point. Therefore, we also attempted genetic algorithms and simulated annealing algorithms to optimize our material selection artificial neural network.

$$x_k = x_{k-1} - \frac{f'(x_{k-1})}{f''(x_{k-1})} \tag{1}$$

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x \partial x} & \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial x \partial z} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y \partial y} & \frac{\partial^2 f}{\partial y \partial z} \\ \frac{\partial^2 f}{\partial z \partial x} & \frac{\partial^2 f}{\partial z \partial y} & \frac{\partial^2 f}{\partial z \partial z} \end{bmatrix} \quad (2)$$

$$B_{k+1} = B_k + \frac{y_k y_k^T}{y_k^T S_k} - \frac{B_k S_k S_k^T B_k}{S_k^T B_k S_k} \quad (3)$$

Genetic algorithms and simulated annealing algorithms are both heuristic algorithms. In this case, the genetic algorithm mimics Darwin's theory of evolution by setting up populations and genotypes for problems that are used to evaluate the fitness function of genotypes. Then, during each iteration, individuals of different genotypes in the population are selected by the fitness function. Individuals with high fitness are retained, while those with low fitness are eliminated. Genetic mutations are then performed, followed by a second iteration until the end condition we set is reached. For our material parameter selection optimization problem, the population is the set of existing material parameter combinations, and each genotype is a material parameter combination. Since our goal is to minimize warpage, our fitness function is shown in (4). Genetic mutation is the change of a part of the material parameters. After iteration, the final optimal genome is the best result of our material selection neural network predicted by the genetic algorithm.

$$Fitness = \frac{1}{1+predict\_value} \quad (4)$$

The simulated annealing algorithm is inspired by the process of metal annealing in solid-state physics. The algorithm gradually reduces the temperature to control the probability of receiving suboptimal solutions during the iteration process, thereby escaping from local optima. For our research problem, we first randomly take a set of material parameter combinations as the initial solution, then randomly change the material parameter combinations to generate the neighborhood solution, and predict whether this neighborhood solution is a more optimal solution through the neural network, if it is, we directly accept the new solution, and if it is not, then we judge whether to accept it or not in accordance with the probability of the equation (5), and then carry out the second round of iteration. Delta means the value of subtract the value of the old solution from the new solution, and current_temp means the temp of this iteration. As the number of iterations increases, the temperature gradually decreases, the results will gradually converge to the global optimal solution to avoid falling into the local optimum.

$$Accept\_prob = e^{-\text{delta}/current\_temp} \quad (5)$$

III. RESULT AND DISCUSSION

In this section, we first demonstrate the effectiveness of FCNN trained on existing data. Then, the ANN was optimized using three different algorithms: BFGS, genetic algorithm, and simulated annealing. Finally, the effectiveness of the three different algorithms was verified, analyzed, and discussed.

A. Artificial Neural Network

We use the grid search method to select and optimize hyperparameters for ANN models. Finally, we constructed an artificial neural network containing three hidden layers, plus the input and output layers in a total of five layers of fully connected structure. Each hidden layer contains 50 neurons, the Rectified Linear Unit (relu) is chosen as the activation function, and the Quasi-Newton methods "lbfgs" are chosen as the solver to optimize the loss function. This model converges after 1099 iterations. The results are shown in Fig. 3 and Table IV.

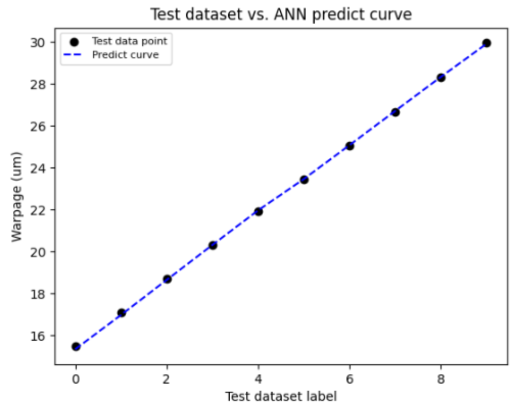

Fig. 3. The ANN predicted data compared to test data of warpage

TABLE IV. RESULT OF ANN MODEL

Evaluation indexes	R2	MSE	MAE	Time (s)
Value	0.9998	0.005	0.05	0.514

B. BFGS Method

We used the minimize function in the scikit—learn [9] package to achieve the effect of the BFGS quasi-Newton method, but since the BFGS parameter cannot constrain the parameter variables, we actually used the L-BFGS-B method to continue the optimization. We also implemented the use of the complete BFGS algorithm to update the Hessian matrix through the trust—constr method in minimize function, while constraining the variable range. However, the actual effect was not as good as the L-BFGS-B method, so we followed the effect achieved by the L-BFGS-B method as the standard. The predicted numerical changes with iteration are shown in Fig. 4, and the best predicted warpage result is 12.367um.

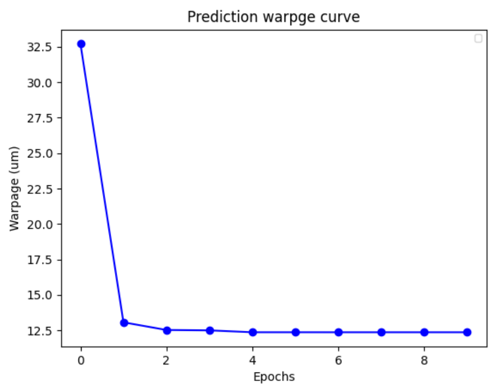

Fig. 4. The BFGS predicted warpage value descent curve

C. Genetic Algorithm

Our genetic algorithm is implemented through the pygad package [10], with 100 iterations and a population size of 100.

The number of genes is 10, which is the same as the number of variables in the ANN model. The proportion of randomly mutated genes is 10%. The fitness improvement curve and warpage prediction curve are shown in Fig. 5, Fig. 6, and the optimal warpage value is 13.575um.

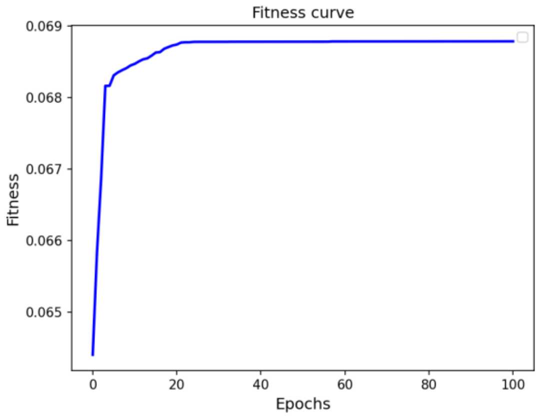

Fig. 5. The genetic algorithm fitness ascend curve

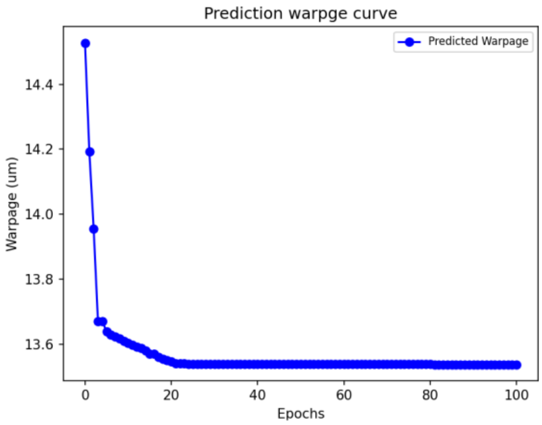

Fig. 6. The genetic algorithm predicted warpage value descent curve

D. Simulated Annealing

We have implemented a simulated annealing algorithm with an initial temperature of 1000 degrees, a cooling rate of 0.995, an iteration number of 3000, and the acceptance probability of the differential solution as shown in (5). The predicted warpage value is shown in figure 7, and the best predicted warpage value is 12.542um.

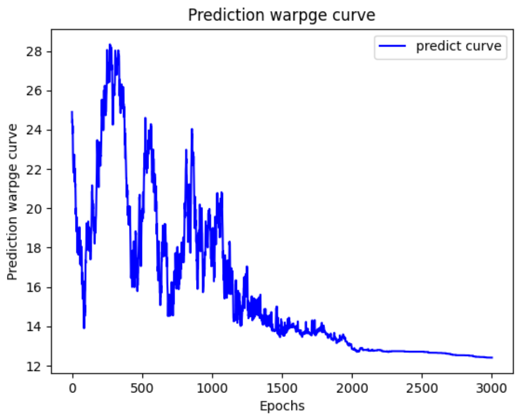

Fig. 7. The simulated annealing algorithm predicted warpage value descent curve

E. Results verification and Algorithm comparison

TABLE V. MATERIAL PARAMETERS AND RESULT TABLE

Model	BFGS	GA	SA
Lid_CTE (ppm/°C)	14	14.57	14
Lid_E (Mpa)	130000	117076	129736
Adhesive_CTE (ppm/°C)	130	130	130
Adhesive_E (Mpa)	7.89	8.008	7.533
Tim_CTE (ppm/°C)	120	131	133.8
Tim_E (Mpa)	50	50	50
Underfill_CTE (ppm/°C)	70	73.8	70
Underfill_E (Mpa)	6000	7221.6	6000
Sub_CTE (ppm/°C)	13	13.05	13
Sub_E (Mpa)	14000	17876	14000
Predicted warpage value (um)	12.4	13.6	12.54
Simulated Warpage Value (um)	12.69	13.7	12.8
Time cost (s)	0.1375	1.8397	0.5227

The optimal material parameter combinations and results obtained by the three algorithms are shown in Table V. We can draw several conclusions from the table. Firstly, the warpage values predicted by the three methods are very close to those verified by FEA, indicating that our ANN model can effectively predict the warpage values during the same process based on material parameters. Secondly, the BFGS method achieved the best results both in terms of results and convergence speed. In terms of convergence speed, the BFGS method, as a numerical solution method, is indeed able to obtain faster results compared to heuristic algorithms such as genetic algorithms and simulated annealing, which corresponds well to the nature of the algorithm itself. As for the results, the BFGS algorithm did not encounter the problem of converging to the local optimum. This is because although our ANN model contains ten variables, the overall nonlinearity is not very strong, which means that there are no particularly complex surfaces in high-dimensional space. Therefore, its local optimum has become its global optimum, and there has been no local optimum problem. It can also be seen from the convergence process that the algorithm predicts values close to 12.5um in almost two steps. Meanwhile, as a numerical solution method, BFGS can achieve higher accuracy in solving the same optimal point than heuristic algorithms such as genetic algorithms and simulated annealing. Thirdly, compared to the other two methods, genetic algorithms yield poorer results and may not be suitable as a commonly used method for such optimization problems. Finally, from the perspective of convergence process, whether it is the fast solution of BFGS method, the Darwin theory of genetic algorithm, simulated annealing jumping out of local optima with temperature and gradually converging to the global optimum, all have achieved the effect of the algorithm, and the time consumption is within 5 seconds. Compared to manually analyzing the interaction effects between different materials, it demonstrates advantages in both time and complexity.

IV. CONCLUSIONS

We constructed a FCBGA finite element simulation model based on an actual chip structure. And a material dataset was constructed within a reasonable material range to address the issue of material selection. A fully connected neural network model was constructed based on this dataset, and three algorithms including BFGS, genetic algorithm, and simulated annealing were used to optimize the neural network and obtain the optimal combination of material parameters. The conclusion is as follows: firstly, ANN neural networks can effectively predict warpage based on material parameters. Secondly, for our simple model with only ten parameters, the BFGS method has demonstrated advantages in both results and convergence speed. Thirdly, for the optimization problem of material selection using ANN, genetic algorithms have achieved poorer results compared to simulated annealing and BFGS algorithms.

It is worth to mentioning that we believe the second conclusion is mainly due to the relatively simple structure of the model. If we increase the number of variables, such as inputting material parameters as a variable every ten degrees with temperature changes, the complexity of the model may increase, and BFGS may exhibit high memory consumption and be prone to getting stuck in local optima. This is also our future research direction.

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